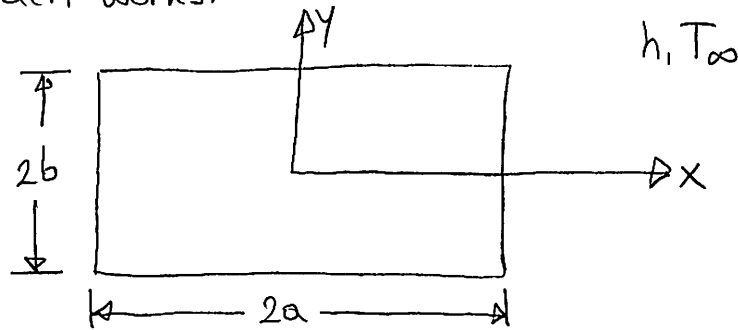


2-D Finite conduction (or 3-D)

Note, if we had a 2-D slab, the same separation of variables approach works:



$$\Theta = f(\bar{x}, \bar{y}, Bi_x, Bi_y, Fo_x, Fo_y)$$

$$\Theta = \Theta_{x,t} \cdot \Theta_{y,t}$$

$$\Theta_{x,t} = f(\bar{x}, Fo_x, Bi_x)$$

$$\bar{x} = \frac{x}{a}$$

$$Fo_x = \frac{\alpha t}{a^2}$$

$$Bi_x = \frac{ha}{k}$$

$$\Theta_{y,t} = f(\bar{y}, Fo_y, Bi_y)$$

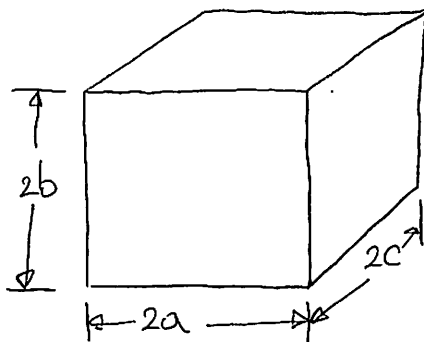
$$\bar{y} = \frac{y}{b}$$

$$Fo_y = \frac{\alpha t}{b^2}$$

$$Bi_y = \frac{hb}{k}$$

Use charts to solve for each direction & then $\Theta = \Theta_{x,t} \cdot \Theta_{y,t}$

For a 3-D body, it would be the same:



$$\Theta = \Theta_{x,t} \cdot \Theta_{y,t} \cdot \Theta_{z,t}$$

Use tabulated results.

Steady Multi-Dimensional Heat Transfer (The Shape factor)

A heat conduction shape factor, S , may be defined for steady problems involving isothermal surfaces as follows:

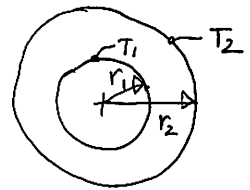
$$Q = Sk\Delta T$$

↳ Obtained analytically or numerically (for complex problems)

$$R = \frac{1}{kS}$$

For example, if we look at a cylinder:

$$S_{cyl} = \frac{1}{kR_{cyl}} \rightarrow R_{cyl} = \frac{|\ln(r_2/r_1)|}{2\pi kL}$$



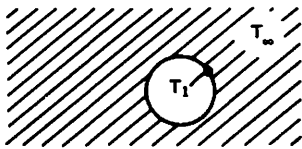
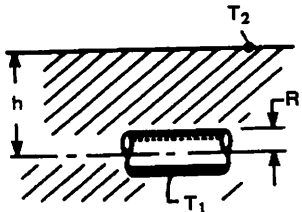
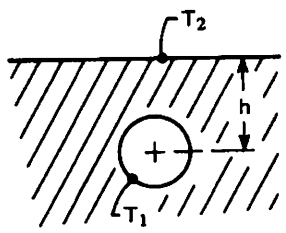
$$S_{cyl} = \frac{2\pi L}{\ln(r_2/r_1)}$$

In general, values are tabulated and solutions are in the form:

$$Q = Sk\Delta T, \quad R_t = \frac{1}{kS}$$

See Mills Table 3.2, page 162-163

Table 5.4 Conduction shape factors: $Q = S k \Delta T$, $R_t = 1/(kS)$.

Situation	Shape factor, S	Dimensions	Source
1. Conduction through a slab	A/L	meter	Example 2.2
2. Conduction through wall of a long thick cylinder	$\frac{2\pi}{\ln(r_o/r_i)}$	none	Example 5.9
3. Conduction through a thick-walled hollow sphere	$\frac{4\pi(r_o r_i)}{r_o - r_i}$	meter	Example 5.10
4. The boundary of a spherical hole of radius R conducting into an infinite medium 	$4\pi R$	meter	Problems 5.19 and 2.15
5. Cylinder of radius R and length L , transferring heat to a parallel isothermal plane; $h \ll L$ 	$\frac{2\pi L}{\cosh^{-1}(h/R)}$	meter	[5.16]
6. Same as item 5, but with $L \rightarrow \infty$ (two-dimensional conduction)	$\frac{2\pi}{\cosh^{-1}(h/R)}$	none	[5.16]
7. An isothermal sphere of radius R transfers heat to an isothermal plane; $R/h < 0.8$ (see item 4) 	$\frac{4\pi R}{1 - R/2h}$	meter	[5.16, 5.17]

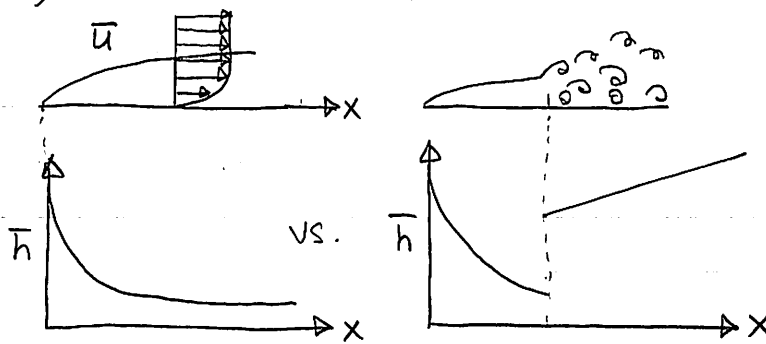
Convection Heat Transfer

Convection is the transfer of thermal energy due to both conduction and by bulk "carrying" of the energy through the velocity of the fluid.

Four Categories:

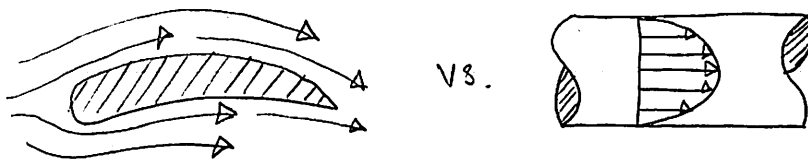
1) Forced vs. Free (Natural)

2) Laminar vs. Turbulent

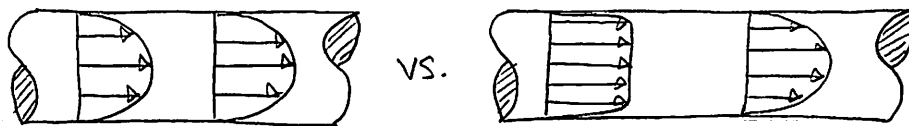


⇒ We will find out why.

3) External vs. Internal

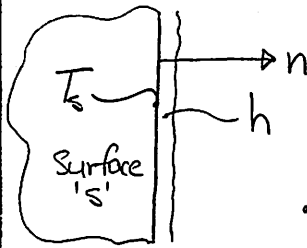


4) Fully Developed vs. Developing



We assume:

- 1) Steady State
- 2) Constant properties
- 3) Incompressible



$$q'' = h(T_s - T_f) \quad (1)$$

$$q'' = -k_f \left. \frac{\partial T}{\partial n} \right|_s \quad (2)$$

where:

h = heat trans. coeff.

T_s = surface temp.

T_f = fluid temp faraway

k_f = fluid thermal cond.

T =

Non-dimensionalizing:

Let:

$$\Theta = \frac{T - T_f}{T_s - T_f}, \quad n^* = \frac{n}{L}$$

Back substitute and equate (1) & (2)

$$-k_f \frac{(T_s - T_f)}{L} \left. \frac{\partial \Theta}{\partial n^*} \right|_s = h(T_s - T_f)$$

$$-\frac{\partial \Theta}{\partial n^*} = \frac{hL}{k_f} = Nu \Rightarrow \text{Nusselt number}$$

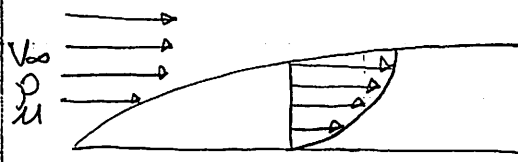
$$Nu = \frac{hL}{k_f} = \frac{\text{convective heat transfer rate}}{\text{conductive heat transfer rate}}$$

Since $Nu \propto \frac{\partial \Theta}{\partial n^*} \propto \Theta$, it typically depends on:

- 1) Flow cond.
- 2) Fluid properties
- 3) Geometry
- 4) Boundary conditions

External Flow

Assuming ρ, μ, C = constant (density, viscosity, heat capacity)
Linked to heat transfer because of the boundary layer



$\Rightarrow$ Where all of the gradients happen in velocity

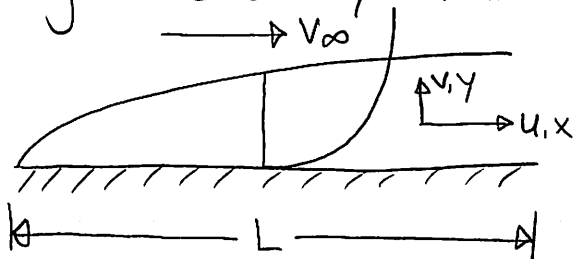
2-D Laminar Flow: (Flat Plate, steady)

x-momentum: $u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right]$

y-momentum: $u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right]$

continuity: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

Looking more closely at our boundary layer:



Let: $\bar{u} = \frac{u}{V_\infty}$, $\bar{v} = \frac{v}{V_\infty}$, $\bar{p} = \frac{p}{\rho V_\infty^2}$

$\bar{x} = \frac{x}{L}$, $\bar{y} = \frac{y}{L}$

Now our equations become:

$\bar{u} \frac{\partial \bar{u}}{\partial \bar{x}} + \bar{v} \frac{\partial \bar{u}}{\partial \bar{y}} = -\frac{\partial \bar{p}}{\partial \bar{x}} + \frac{1}{Re_L} \left[\frac{\partial^2 \bar{u}}{\partial \bar{x}^2} + \frac{\partial^2 \bar{u}}{\partial \bar{y}^2} \right]$, $Re_L = \frac{\rho V_\infty L}{\mu}$

$\bar{u} \frac{\partial \bar{v}}{\partial \bar{x}} + \bar{v} \frac{\partial \bar{v}}{\partial \bar{y}} = -\frac{\partial \bar{p}}{\partial \bar{y}} + \frac{1}{Re_L} \left[\frac{\partial^2 \bar{v}}{\partial \bar{x}^2} + \frac{\partial^2 \bar{v}}{\partial \bar{y}^2} \right]$

$\frac{\partial \bar{u}}{\partial \bar{x}} + \frac{\partial \bar{v}}{\partial \bar{y}} = 0$

So what can we say about these equations and their relative terms? $u \sim V_\infty$, $x \sim L$, $y \sim \delta$ (boundary layer thick.)

$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{V_\infty}{L} + \frac{v}{\delta} = 0 \Rightarrow v \sim \left(\frac{\delta}{L} \right) V_\infty$

Looking back at our momentum equations

↳ Note dropped the negative sign since scaling analysis.

Going to x-momentum

$$\underbrace{V_\infty \frac{V_\infty}{L} + V_\infty \left(\frac{\delta}{L}\right) \cdot \frac{V_\infty}{\delta}}_{\sim \frac{V_\infty^2}{L}} = \underbrace{-\frac{1}{\rho} \frac{\partial p}{\partial x}}_{\sim \frac{V_\infty^2}{L}} + \underbrace{\nu \left[\frac{V_\infty}{L^2} + \frac{V_\infty}{\delta^2} \right]}_{\sim \frac{\nu V_\infty}{\delta^2}}$$

≈ 0 since $\frac{\delta}{L} \ll 1$

Can rewrite as:

$$\frac{V_\infty}{L^2} = \frac{\delta^2}{L^2} \cdot \frac{V_\infty}{\delta^2} \approx 0 \text{ since } \frac{\delta}{L} \ll 1$$

y-momentum:

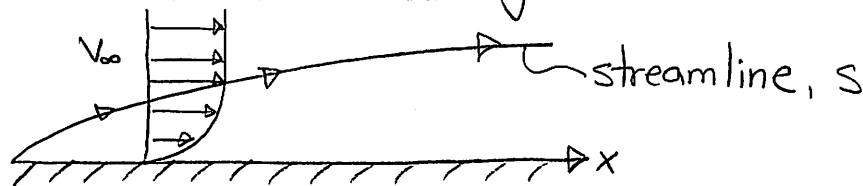
$$\underbrace{V_\infty \frac{\delta}{L} V_\infty \cdot \frac{1}{L}}_{\sim \frac{V_\infty^2}{L} \cdot \left(\frac{\delta}{L}\right)} + \underbrace{V_\infty^2 \left(\frac{\delta}{L}\right)^2 \frac{1}{\delta}}_{\sim \frac{V_\infty^2}{L} \cdot \left(\frac{\delta}{L}\right)} = \underbrace{-\frac{1}{\rho} \frac{\partial p}{\partial y}}_{\sim \frac{V_\infty^2}{\delta}} + \underbrace{\nu \frac{\delta}{L} V_\infty \left[\frac{1}{L^2} + \frac{1}{\delta^2} \right]}_{\sim \frac{\nu V_\infty}{\delta} \left[\frac{1}{L^2} + \frac{1}{\delta^2} \right]}$$

Since $\frac{\delta}{L} \ll 1$, we can drop the inertia terms, & viscous terms

Our y-momentum equation becomes:

$$\frac{\partial p}{\partial y} = 0 \Rightarrow \boxed{p(y) = \text{Constant}} \Rightarrow \text{Pressure is not a function of } y$$

If we look at the edge of our boundary layer



Our x-momentum equation along the streamline s becomes

$$u \frac{\partial u}{\partial x} + \underbrace{v \frac{\partial u}{\partial y}}_{0 \text{ at } s} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \underbrace{\nu \left(\frac{\partial^2 u}{\partial y^2} \right)}_{0 \text{ at } s}$$

$$V_\infty \frac{\partial V_\infty}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = 0 \Rightarrow \boxed{\rho + \frac{1}{2} \rho V_\infty^2 = \text{constant}}$$

↳ Bernoulli equation

Note from this, we can see that the pressure in the x-direction is constant ($P \neq f(x)$) for steady flow past a flat plate, since $V_\infty = \text{constant}$ at the b.l. edge. Note, not true for cylinder, sphere, etc., where V_∞ changes.

$$\frac{\partial V_x}{\partial x} = 0 \quad \text{or} \quad \frac{1}{\rho} \frac{\partial p}{\partial x} = -V_\infty \frac{\partial V_\infty}{\partial x} = 0$$

So our boundary layer equations become:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} \quad (1)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (2)$$

$$p(y) = \text{constant} \quad (3)$$

Boundary conditions:

$$u = v = 0 \quad @ \quad y = 0$$

$$u = U_\infty \quad @ \quad y \rightarrow \infty$$

So now we must convert our PDE into a solvable ODE. Cannot use separation of variables (infinite medium). Let's try a similarity solution.

Assume: $\frac{u}{V_\infty} = f(\eta)$ where $\eta = \frac{y}{\delta}$

So how do we define δ ?

Looking back at our non-dimensional x-momentum:

$$\underbrace{\bar{u} \frac{\partial \bar{u}}{\partial \bar{x}}}_{\sim O(1)} + \underbrace{\bar{v} \frac{\partial \bar{u}}{\partial \bar{y}}}_{\sim O(1)} = \underbrace{\frac{1}{Re_L} \frac{\partial^2 \bar{u}}{\partial \bar{y}^2}}_{\text{Has to be on order of 1}}, \quad \bar{u} = \frac{u}{V_\infty} \sim \frac{V_\infty}{V_\infty} \sim O(1)$$

$$\bar{x} = \frac{x}{L} \sim \frac{L}{L} \sim O(1)$$

$$\frac{1}{Re_L} \frac{V_\infty}{V_\infty} \cdot \left(\frac{\delta}{L}\right)^2 \sim 1$$

$$\bar{y} = \frac{y}{L} \sim \frac{\delta}{L}$$

$$\frac{1}{Re_L} \frac{\delta^2}{L^2} \sim 1 \Rightarrow \frac{\delta}{L} \sim \frac{1}{\sqrt{Re_L}} \quad \text{or} \quad \frac{\delta}{x} \sim \frac{1}{\sqrt{Re_x}}$$

$$\boxed{\delta \sim \sqrt{\frac{\nu x}{V_\infty}}}$$

so now we get:

$$\boxed{\eta = y \sqrt{\frac{V_\infty}{\nu x}}}$$

↳ Will see an easier way to get this later.

So what about v ? Looking at continuity:

$$\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} = 0 \Rightarrow \bar{u} = \phi(\eta) = \frac{u}{V_\infty}$$

$$\frac{\partial \bar{u}}{\partial x} = \frac{\partial \bar{u}}{\partial \eta} \cdot \frac{\partial \eta}{\partial x}$$

$$\frac{\partial \bar{u}}{\partial x} = -\frac{1}{2x} \eta \phi'$$

Back substituting, we obtain:

$$\frac{\partial \bar{v}}{\partial y} = \frac{1}{2x} \phi' \eta$$

$$\bar{v} = \frac{1}{2x} \int_0^\eta \phi' \eta \, d\eta = \underbrace{\frac{1}{2x} \sqrt{\frac{xV}{V_\infty}}}_{\frac{1}{2} \frac{1}{\sqrt{Re_x}}} \int_0^\eta \phi' \eta \, d\eta$$

Aside:

We know $\eta = y \sqrt{\frac{V_\infty}{\nu x}}$
 $d\eta = dy \sqrt{\frac{V_\infty}{\nu x}}$

Now we need to solve our integral:

$$\int_0^\eta \eta \phi' \, d\eta = u v - \int v \, du \quad (\text{Integration by parts: } \int u \, dv = uv - \int v \, du)$$

$$= \eta \phi - \int_0^\eta \phi \, d\eta, \quad u = \eta, \quad v = \phi$$

$$\text{let } \int_0^\eta \phi \, d\eta = F(\eta)$$

$$\text{So now we have: } \bar{v} = \frac{1}{2} \frac{1}{\sqrt{Re_x}} (\eta F' - F)$$

Putting everything together we get:

$$F''' + \frac{1}{2} F F'' = 0; \quad \eta = y \sqrt{\frac{V_\infty}{\nu x}}$$

At the wall: $\eta = 0, \quad F' = F = 0$
 Outside the b.l: $\eta \rightarrow \infty, \quad F' = 1$

Rewriting our equation:

$$\frac{\partial^2 \bar{u}}{\partial \eta^2} + \frac{1}{2} F \frac{\partial \bar{u}}{\partial \eta} = 0$$

To solve, we need to assume an infinite series solution

$$F = a_0 + a_1 \eta + a_2 \eta^2 + a_3 \eta^3 + \dots +$$

$$F' = a_1 + 2a_2 \eta + 3a_3 \eta^2 + \dots +$$

$$F'' = 2a_2 + 6a_3 \eta + \dots +$$

$$F''' = 6a_3 + \dots +$$

Back substitute and solve for your coefficients

$$F''' + \frac{1}{2} F F'' = 0$$

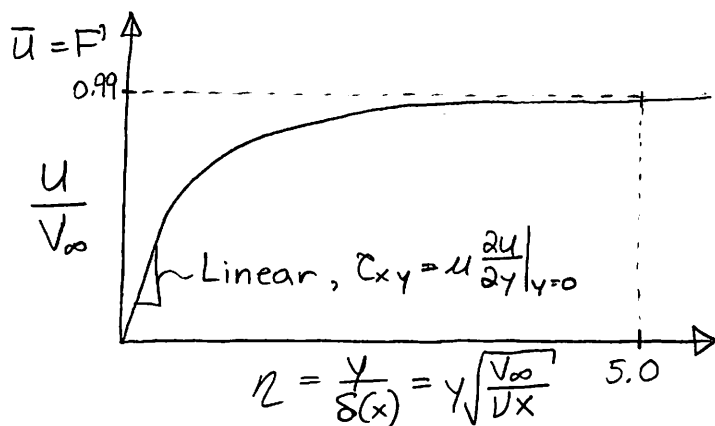
$$\underbrace{(\dots)}_0 \eta^0 + \underbrace{(\dots)}_0 \eta^1 + \underbrace{(\dots)}_0 \eta^2 + \underbrace{(\dots)}_0 \eta^3 + \dots + \underbrace{(\dots)}_0 \eta^n = 0$$

You obtain a recursion formula relating your constants

$$F = \frac{a_2 \eta^2}{2!} - \frac{a_2 \eta^5}{2 \cdot 5!} + \frac{11}{4} \frac{a_3^3 \eta^8}{8!} + \dots$$

$a_2 = 0.332 \Rightarrow$ Heinrich Blasius solved this in 1911 for his Ph.D. work with Ludwig Prandtl.

$$\text{So } F = \int_0^\eta \phi d\eta, \quad F' = \phi = \bar{u} = \frac{u}{V_\infty}$$



$$5.0 = \delta \sqrt{\frac{V_\infty}{\nu x}} = \frac{\delta}{x} \sqrt{Re_x}$$

$$\delta = \frac{5x}{\sqrt{Re_x}}$$

$\Rightarrow$ Hydrodynamic boundary layer thickness.

This is very usefull because now we can calculate things like shear stress

$$\begin{aligned}\tau(x) &= \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \rho \nu V_{\infty} \left. \frac{\partial \bar{u}}{\partial y} \right|_{y=0} \\ &= \rho \nu V_{\infty} \left. \frac{\partial \bar{u}}{\partial \eta} \right|_{\eta=0} \cdot \left(\frac{\partial \eta}{\partial y} \right) \\ &= \rho \nu V_{\infty} a_2 \sqrt{\frac{V_{\infty}}{x \nu}}\end{aligned}$$

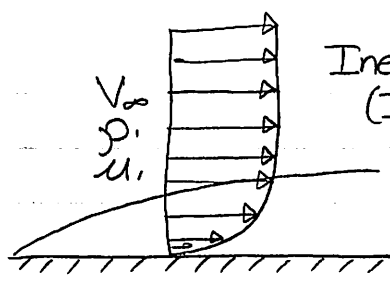
$$\boxed{\frac{\tau(x)}{\frac{1}{2} \rho V_{\infty}^2} = \frac{2a_2}{\sqrt{Re_x}} = \frac{0.664}{Re_x^{1/2}}} \Rightarrow \text{Skin friction coefficient for a flat plate in laminar flow conditions.}$$

If we want the average:

$$\begin{aligned}\bar{\tau} &= \frac{1}{L} \int_0^L \tau(x) dx \Rightarrow \tau(x) = C \frac{1}{\sqrt{x}} \\ &= C \cdot \frac{1}{L} \int_0^L \frac{dx}{\sqrt{x}} = \frac{2C}{\sqrt{L}}\end{aligned}$$

$$\boxed{C_L = \frac{\bar{\tau}}{\frac{1}{2} \rho V_{\infty}^2} = \frac{1.328}{Re_L^{1/2}}} \Rightarrow \text{Average skin friction coeff.}$$

So is there an easier way to see some simple things about boundary layers?



Inertia Dominates (Inviscid Flow) $\Rightarrow u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x}$

} Inside the b.l. Inertia $\sim$ Viscosity
Viscous effects become important.

$$\rho \frac{V_{\infty}^2}{L} \sim \mu \frac{V_{\infty}}{\delta^2} \Rightarrow \delta^2 = \frac{\mu L}{\rho V_{\infty}} \Rightarrow \delta \sim \sqrt{\frac{\nu L}{V_{\infty}}} \left(\frac{L}{L} \right)$$

$$\boxed{\delta \sim \frac{L}{\sqrt{Re_L}}}$$